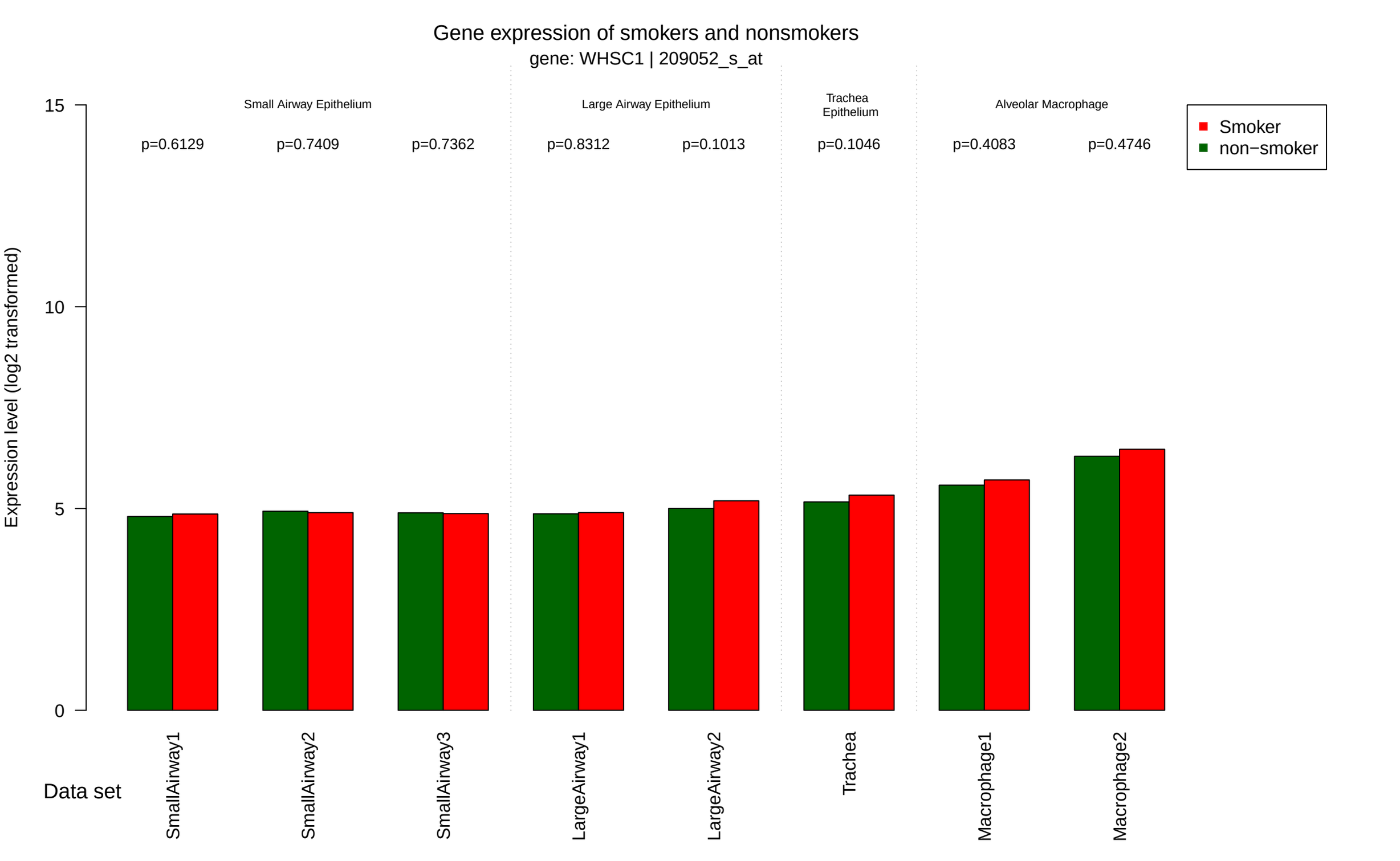
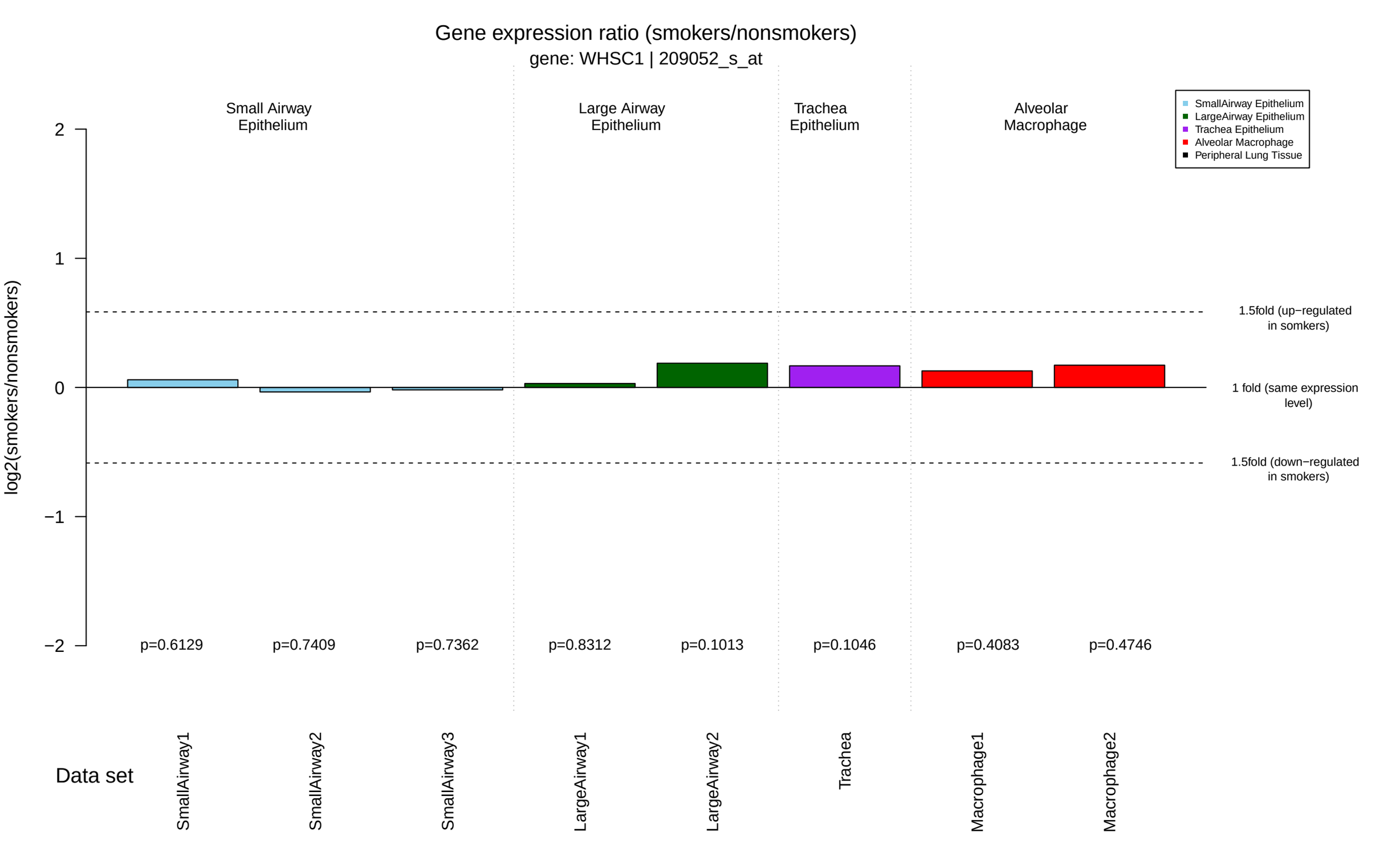
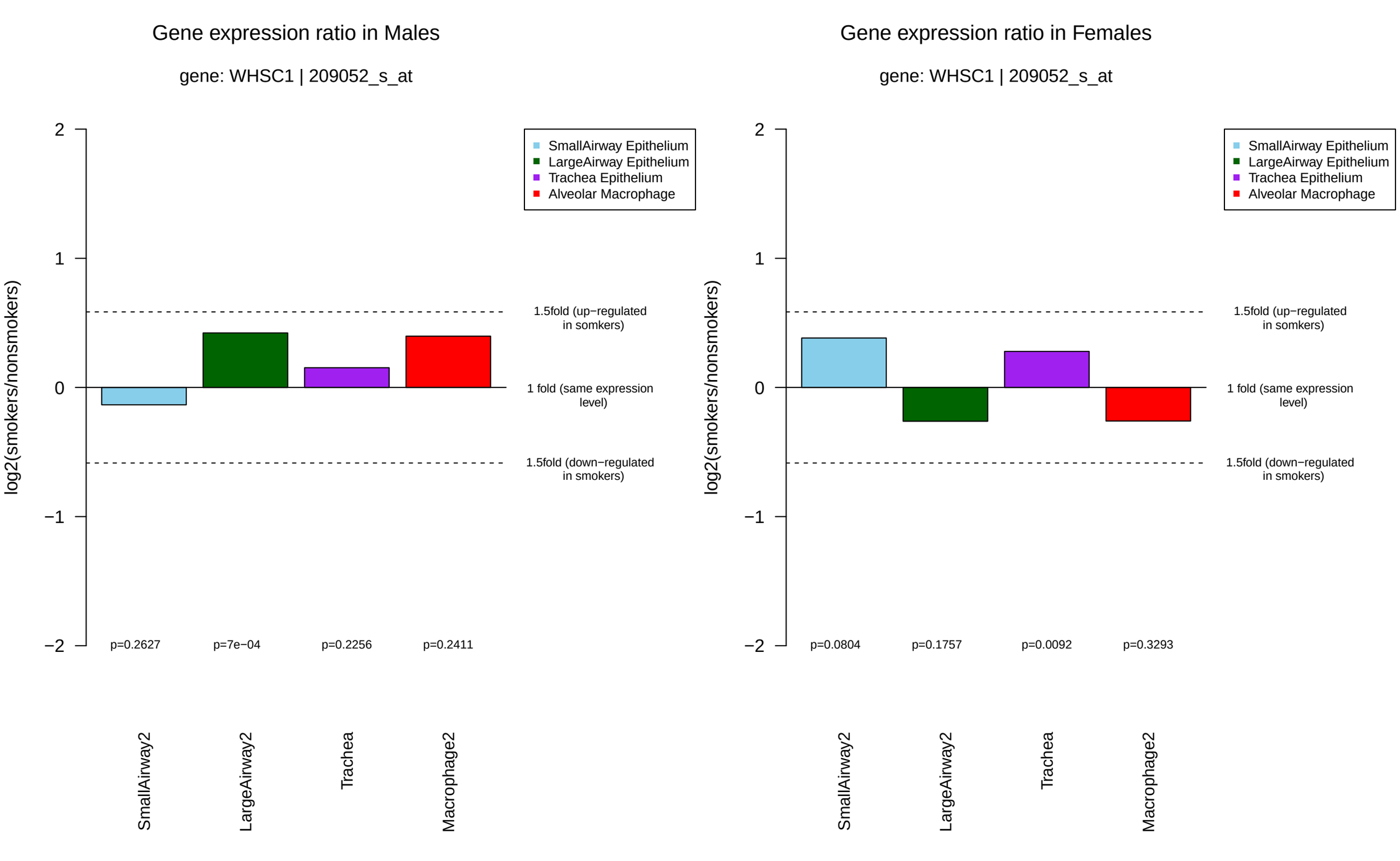
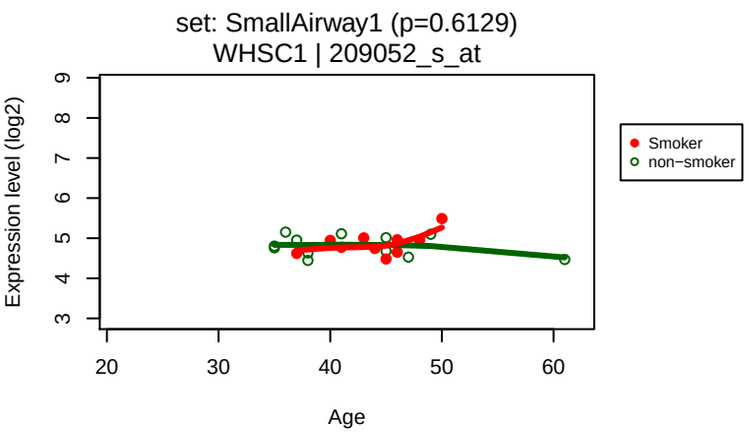
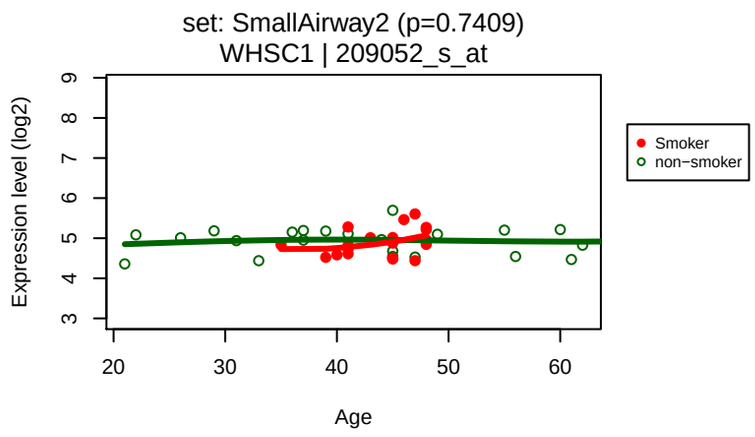
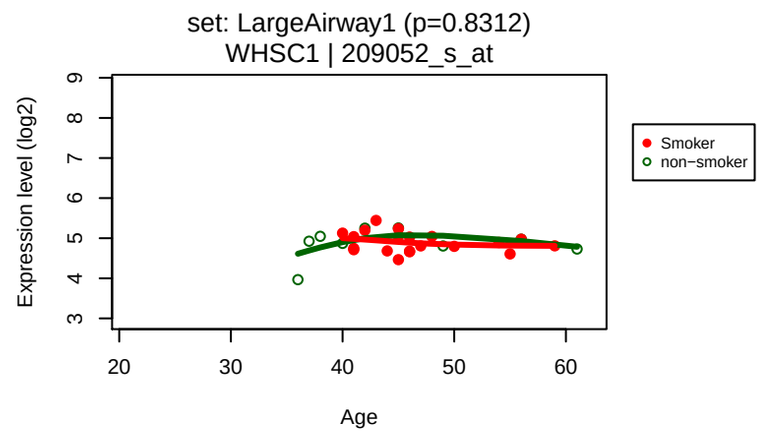

Gene expression of smokers and nonsmokers
gene: WHSC1 | 209052_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6129
p=0.7409
p=0.7362
p=0.8312
p=0.1013
p=0.1046
p=0.4083
p=0.4746
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

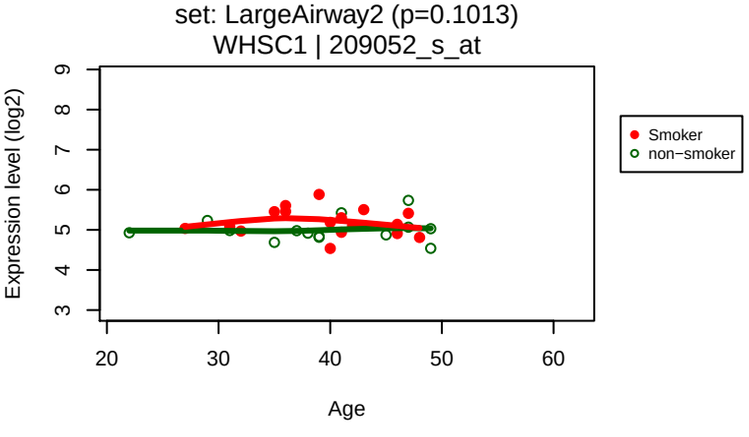

Gene expression ratio (smokers/nonsmokers)
gene: WHSC1 | 209052_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6129
p=0.7409
p=0.7362
p=0.8312
p=0.1013
p=0.1046
p=0.4083
p=0.4746
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

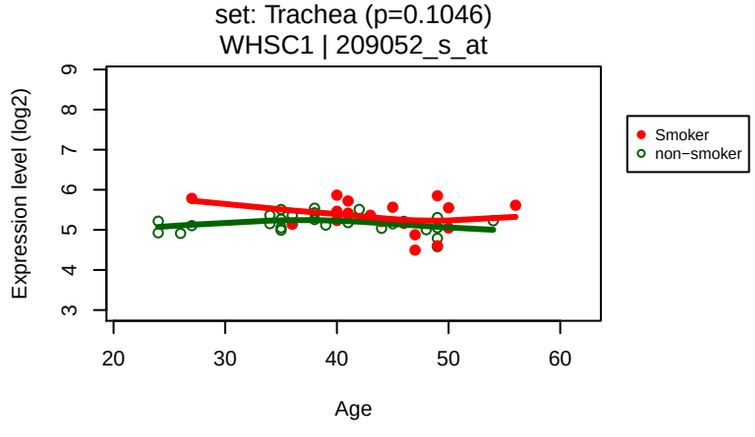

Gene expression ratio in Males
gene: WHSC1 | 209052_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2627
p=7e−04
p=0.2256
p=0.2411
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WHSC1 | 209052_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0804
p=0.1757
p=0.0092
p=0.3293
SmallAirway2
LargeAirway2
Trachea
Macrophage2

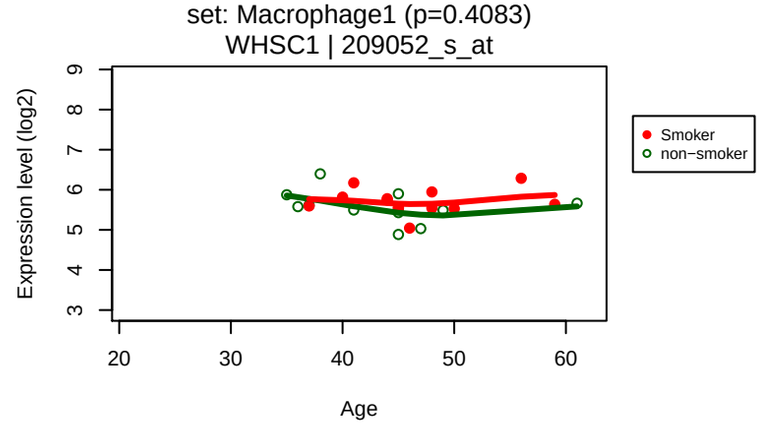

set: SmallAirway1 (p=0.6129)
WHSC1 | 209052_s_at
Expression level (log2)
Age
Smoker
non−smoker

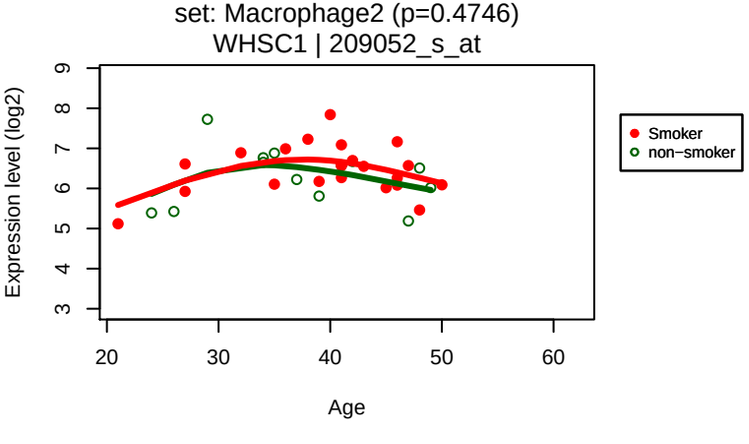

set: SmallAirway2 (p=0.7409)
WHSC1 | 209052_s_at
Expression level (log2)
Age
Smoker
non−smoker

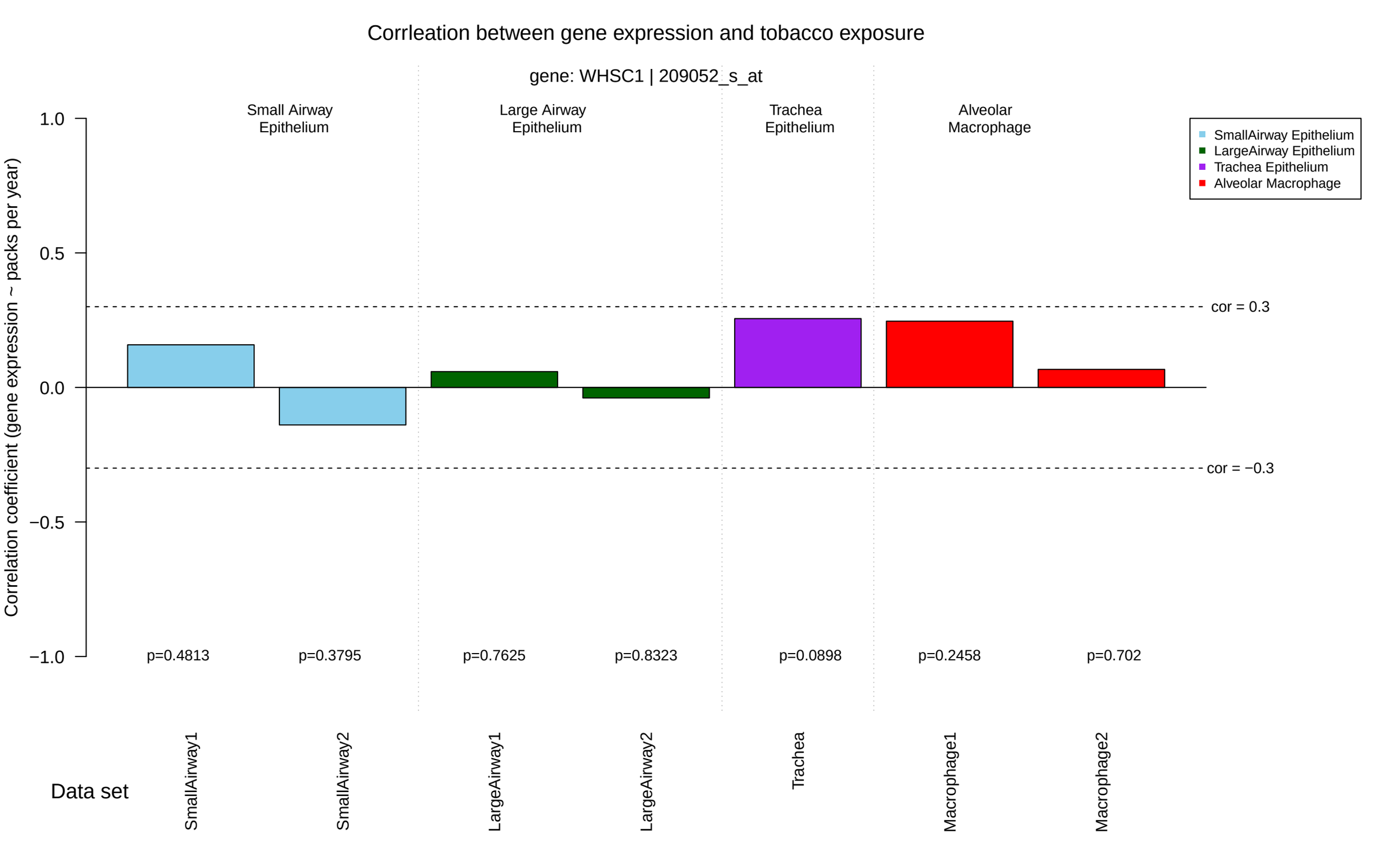

set: LargeAirway1 (p=0.8312)
WHSC1 | 209052_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1013)
WHSC1 | 209052_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1046)
WHSC1 | 209052_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4083)
WHSC1 | 209052_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4746)
WHSC1 | 209052_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WHSC1 | 209052_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4813
p=0.3795
p=0.7625
p=0.8323
p=0.0898
p=0.2458
p=0.702
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2